DysLex-Probe — Defense Speech

Probing Cross-Lingual Lexical Representations in Multilingual LLMs for Low-Resource Dyslexia Screening

Four-person presentation script · 15 slides · approx. 13 minutes · spoken in "we" form

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Slide 1 — Title

~40 sec

Good morning everyone, and thank you for being here. We are [team members' names], and our thesis is called DysLex-Probe. In simple words, we studied whether large multilingual AI language models can help us screen for dyslexia, especially in a low-resource language like Bangla. Over the next thirteen minutes, we will walk you through why this problem matters, the dataset and models we used, the questions we asked, and what we found. Let's begin.

Slide 2 — Motivation

~70 sec

First, why this matters. Dyslexia is a very common reading difficulty. It is important to understand what it is not. It is not about how clever a child is, and it is not an eyesight problem. It is mainly about how the brain handles the sounds of language, what we call phonology. Now, for English, there are proper, well-tested screening tools that doctors and teachers already trust. But for Bangla, a language spoken by more than 270 million people, there is no validated automatic screener at all. And this missing tool is itself a barrier, because a child who is never tested is a child who never gets help. So we asked a simple but important question. The multilingual AI models that have already learned Bangla, do they carry any real knowledge about sound structure that we could borrow for screening? That question is the heart of our work.

Slide 3 — Gap and Research Questions

~60 sec

This slide makes the gap clear. We have good tools for English, almost nothing for Bangla, and nobody has really checked whether today's multilingual models truly understand Bangla

phonology, or whether they just copy simple surface patterns. From this gap, we built four research questions. One, do the models really encode sound knowledge, or are they using cheap tricks? Two, does what a model learns transfer from English to Bangla? Three, where inside the model is this knowledge stored? And four, which model should someone actually choose? These four questions guide the rest of the talk.

Slide 4 — Literature Review

~55 sec

Before building anything, we studied the existing work. On the clinical side, there are standardized tests like CTOPP-2 for English that measure phonological skills. On the AI side, researchers have begun probing language models to see what knowledge sits inside them, but mostly for high-resource languages. For Bangla, the most useful source was Badsha's 2024 study from the University of Dhaka, which recorded the real reading difficulties of dyslexic Bangla-speaking children, like confusing voiced and unvoiced sounds and struggling with conjunct letters. So the picture is clear: plenty of work on English, some probing of models, but very little linking multilingual models to Bangla screening. That link is exactly what we set out to build.

Slide 5 — Approach

~55 sec

Here is our whole approach in one picture. Instead of picking models at random, we chose four models that map out the entire design space. The horizontal axis is the training objective, how the model was taught: masked encoder, causal decoder, or encoder-decoder. The vertical axis is language coverage, from strongly multilingual with Bangla at the top, to more English-heavy at the bottom. Each card shows the model, its size, and the verdict we reached. By spreading our four models across this map, we can fairly ask which design choices really matter for Bangla, instead of just comparing two random systems.

Slide 6 — Dataset

~70 sec

To test the models, we built a dataset of 440 carefully designed items: 220 in English and 220 in Bangla, so the two languages are perfectly balanced. The items fall into seven categories. Minimal pairs, which are words that differ by just one sound. Multisyllabic words.

Pseudowords, which are fake but pronounceable. Real words. Phoneme-to-grapheme items. Rhymes. And full sentences. Each item also carries detailed labels, eighteen columns in total, covering things like syllable count, whether it has a consonant cluster, and the exact dyslexia error it is meant to probe. The Bangla items were designed around the real difficulty patterns from Badsha's study, and then checked by a native speaker to remove any accidental real words. So this is not a random word list. Every single item is there for a reason.

Slide 7 — Models Compared

~55 sec

These are the four models in detail. XLM-R is an encoder-only model, the smallest at about half a billion parameters, and it has Bangla in its training data. BLOOM is a decoder-only model, around seven billion parameters, also with Bangla. mT5 is an encoder-decoder, about three to four billion, again with Bangla. And Llama is a decoder-only model, around eight billion, but trained on very little Bangla. Notice they differ in two big ways: how they were trained, and how much Bangla they have actually seen. This is deliberate, because it lets us separate the effect of model design from the effect of pure size. That separation is the key idea behind the whole comparison.

Slide 8 — Methodology

~60 sec

This slide shows our pipeline from start to finish. For every model, we take each of the 440 items, feed it through the frozen model, and pull out the model's internal representation, basically the numbers the model uses to represent that word. Then we train a very simple classifier on the English representations and ask: how well does it work on Bangla? But the most important part is the honesty checks. We ran three sanity tests to make sure the model was doing real language work and not cheating with shortcuts like word length or the first letter. And finally, we used a method called causal ablation, where we switch off small parts of the model on purpose and watch whether Bangla performance drops. That tells us where the useful knowledge actually lives.

Slide 9 — Research Question 1 Results

~75 sec

Now the findings, and this first one is the most important lesson of the whole thesis. We tested

four standard probing tasks, and three of the four failed our sanity checks. The clearest example is the voicing task, telling voiced sounds from voiceless ones. It turned out the answer could be guessed just from the first letter of the Bangla word. In fact a probe using only the first character scored between 0.78 and 0.94 across all four models. The AI was not doing any sound work at all, it was just reading letters. The spelling and syllable tasks failed too, because simple features like word length already solved them. Only one task survived: telling a real word from a fake one, and only after we carefully matched real and fake words to the same length. On that honest task, every model gained between 0.20 and 0.30 in Bangla over the shortcut baseline. Catching this stopped us from reporting a false success, and that honesty is itself part of our contribution.

Slide 10 — Research Question 2 Results

~70 sec

The second question is about transfer across languages: does knowledge learned in English help in Bangla? Here the results split the four models cleanly in two. On the honest, length-matched real-versus-fake word task, BLOOM scored 0.92 on Bangla and XLM-R scored 0.86, both clearly beating the shortcut baseline, by 0.30 and 0.24. So those two genuinely transfer. But Llama scored only 0.63, and mT5 only 0.60. Both collapsed down to the shortcut level, meaning no real transfer. Now look at which models won. The two that work both have Bangla in their training data. And here is the striking part: XLM-R, at just half a billion parameters, beats Llama at eight billion. So having Bangla in the training data matters far more than having a bigger model. Coverage beats scale.

Slide 11 — Research Question 3 Results

~65 sec

The third question asks where inside the model this knowledge lives, and this is where one model stood out. When we switched off a small set of units inside mT5, its Bangla performance dropped by a clean, measurable amount, about 0.11. When we did the exact same thing to the other three models, the drop was either zero or random. That means mT5 stores its cross-language knowledge in a small, specific set of locations, while the others spread it out everywhere. We then repeated this at five different depths inside mT5, and the pattern held at every depth, strongest in the middle of the encoder, around layer twelve. Two units in particular, numbered 782 and 1880, kept showing up again and again, like a backbone. This is exactly why mT5 earned the verdict "interpretable": it is the only model where we can actually

point to where the knowledge sits, even though its raw transfer is weak.

Slide 12 — Research Question 4 Recommendation

~65 sec

This brings everything together into a practical recommendation. If you want the best Bangla performance for a real screening prototype, use BLOOM, our top transfer model at 0.92. If you want the most efficient option, especially on limited hardware, use XLM-R: it nearly matches the giants at one thirteenth of the size. If you want to understand and inspect how the model represents language, use mT5, the only interpretable one. And for Bangla, do not use Llama: despite being the largest, its cross-lingual transfer collapses to 0.63. The single biggest takeaway is this: the training objective and the language coverage, not the raw size of the model, are what decide cross-lingual transfer. Bigger is not automatically better.

Slide 13 — Future Work

~45 sec

There are clear next steps. First, the dataset can grow, with more Bangla items and more dialects, since we only covered standard forms. Second, and most importantly, the screening idea should now be tested with real children, alongside clinical experts, to move from probing inside a model to a tool people can actually use. Third, the backbone-units finding in mT5 was striking, but mT5 was the only encoder-decoder we tested, so this should be checked on more models. And finally, our sanity-check lesson means future tasks must be designed so that models cannot shortcut them.

Slide 14 — Conclusion

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To conclude. We began with a real gap: no automated dyslexia screener exists for Bangla. We built a balanced, carefully designed 440-item dataset, and tested four very different multilingual models on it. Our biggest finding is a warning: three out of four standard probing tasks were measuring shortcuts, not real understanding. Once we fixed that, BLOOM and XLM-R clearly carried genuine knowledge from English to Bangla, mT5 turned out to be the only model where we can see how that knowledge is stored, and the deciding factor was language coverage, not size. We also want to be clear: this is a research probe, not a clinical diagnostic tool. But it is a solid first step toward fair, automated dyslexia screening for a low-

resource language.

Slide 15 — Thank You and Questions

~20 sec

That brings us to the end of our presentation. Thank you all very much for your time and your attention. We would be very happy to take any questions you may have.

Total running time: approximately 13 minutes at a calm pace (~140 words per minute).

> **Notes for the team.** All result figures in this script come from the project's own results files (comparison_four_models.md / faculty report): voicing first-character F1 0.78–0.94; real-vs-pseudo gains +0.20 to +0.30; cross-lingual transfer BLOOM 0.92, XLM-R 0.86, Llama 0.63, mT5 0.60; mT5 ablation specificity +0.11 peaking at layer 12; backbone units 782 and 1880. Fill in the four presenter names on Slide 1.